

OINDRILA SAHA BADHAN

 Portfolio |  Github |  Oindrila Saha |  Google Scholar |  oindrilasaha.001@gmail.com |  Codeforces |  +880 1315622658 |

1500 W Plum Street, Fort Collins, Colorado 80521

OBJECTIVE

AI researcher with interests spanning **Computer Vision, NLP, LLMs, XAI**, and **Human-Computer Interaction (HCI)**. Passionate about developing interpretable and efficient AI systems that bridge human interaction and intelligent technologies for real-world healthcare and research applications.

EDUCATION

- Ahsanullah University of Science and Technology** 2019 - 2023
Bachelor of Science (B.Sc.), Computer Science and Engineering **GPA : 3.20 / 4.00**
- Holy Cross College** 2016 - 2018
Higher Secondary Certificate
- Kamrunnesa Govt Girls High School** 2009 - 2016
Secondary School Certificate

ENGLISH PROFICIENCY

IELTS Academic: Overall 7.5 (Listening: 8.0, Reading: 8.0, Writing: 6.5 , Speaking: 7.0)

RELEVANT COURSES

- Artificial Intelligence
- Soft Computing
- Algorithms
- Digital Image Processing
- Mathematical Analysis for Computer Science
- Pattern Recognition
- Operating System
- Distributed Database Systems
- Data Structures
- Discrete Mathematics
- Computer Graphics
- Computer Networks
- Microprocessors
- Computer Architecture

RESEARCH EXPERIENCE

- Research Collaborator (Supervisor: Khan Md Hasib)** Sept 2026 - Present
 - Constructed a machine learning dataset from the MIMIC-III clinical database by integrating discharge summaries with multi-label ICD-9 diagnoses using PostgreSQL and SQL.
 - Investigating a hybrid LLM-GNN architecture to jointly model clinical text semantics and disease co-occurrence relationships for multi-label disease prediction.
- Research Collaborator, Computer Vision & AI Lab (Supervisor: Dr. Mohammad Shafiul Alam)** June 2026 - Present

Working as **1st author** on multiclass disease detection of lemon leaves using deep learning architectures spanning diverse paradigms. Implementing **lightweight (MobileNetV3)**, **deep residual (ResNet101V2)**, and **densely connected (DenseNet121)** networks to optimize feature extraction and classification accuracy.

 - * Exploring **multi-scale models (InceptionV3, Xception, Inception-ResNetV2)** and **efficient scaling architectures (EfficientNetB0, EfficientNetB3, EfficientNetV2-S)** for balancing computational cost and performance.
 - * Evaluating **modern CNN (ConvNeXt-Small)** and **transformer-based (EfficientViT)** designs to capture global and local leaf disease patterns, aiming to inform real-time agricultural disease monitoring.
 - * Applying **explainable AI (XAI)** techniques such as Grad-CAM to interpret model predictions, while benchmarking performance using accuracy, precision, recall, and F1-score to compare efficiency, scalability, and practical deployment suitability
- Research Collaborator, AI Lab (Supervisor: Mr. Nazmus Sakib)** 2024 - 2025
 - * As **1st author** conducted a **review paper** presenting a **PRISMA-based systematic review** of road crack detection using deep learning-based **lightweight models**. Analyzed over **100 research papers**, evaluating **model architectures, strengths, weaknesses, and future directions**.
 - * Examined practical deployment strategies and **provided guidance on selecting attention mechanisms, loss functions, and model variants** for architectures such as **YOLO, U-Net, MobileNet, transformer, and hybrid models**.
 - * Conducted **comparative performance analysis** to identify the best-performing variants and architecture designs for different scenarios.

- * Synthesized insights to **guide future research and development** of efficient, scalable, and deployable lightweight models for real-world infrastructure monitoring.
- **Research Collaborator at NSU Applied ML/DL Lab** 2024-2025
 - * Explored CNN and transfer learning architectures (VGG16, ResNet50, EfficientNet, MobileNetV2) for automated soft tissue profile classification, enhancing model performance through dataset balancing and GAN-based augmentation.
- **B.Sc. Thesis(Supervisor: Ms. Syeda Shabnam Hasan)** 2022-2023
 - * My undergraduate thesis focused on emotion detection from Bengali text, addressing the scarcity of NLP resources for low-resource languages. I developed a novel Bengali emotion dataset comprising 4,300 sentences categorized into six emotions — happiness, sadness, fear, anger, disgust, and surprise. Various deep learning and transformer-based models (**LSTM, GRU, CNN hybrids, BanglaBERT, mBERT, DistilBERT**) were systematically evaluated. The proposed CNN+GRU hybrid model achieved 92.28% accuracy, outperforming traditional architectures and demonstrating the potential of hybrid deep learning approaches for nuanced emotion recognition in Bengali language contexts.

RESEARCH WORKS

- **Published Work**
 - * "Emotion Recognition in Bengali Text: A Computational Approach for Six Emotions," *IEEE International Conference on Computer and Information Technology (ICCIT)*, 2025. DOI: [10.1109/ICCIT68739.2025.11491714](https://doi.org/10.1109/ICCIT68739.2025.11491714)
- * **Journal Work**
 - "Lightweight deep learning architectures for real-time road crack detection: A systematic review and performance analysis", (Under Review, Q1/Q2 journal), 2025. **[1st Author]**
 - "Hybrid LLM-GNN Framework for Multi-label Disease Prediction using MIMIC-III ", (Ongoing), 2026.
- * **Conference Works**
 - "CNN-Based Automated Diagnosis of Lemon Leaf Diseases", (Ongoing) **[1st Author]**
 - "Deep Learning-Based Classification of Soft Tissue Facial Profiles in Orthodontics", (Ongoing) **[1st Author]**

NOTABLE ACADEMIC PROJECTS

- * **3D Game Development | Unity, C#** | Developed an interactive 3D game using Unity, implementing player controls, gameplay mechanics, physics, interactive environments, and real-time user interaction.
- * **University Management System (UniCentral) | Java, Swing, MySQL, Java 8** | Developed a centralized platform for academic and administrative management, featuring admin login, student & teacher management, exam & marks handling, inventory tracking, search functionality, report generation, and integrated calculator for essential computations. [\[Github Link\]](#)
- * **Art Gallery Management System | Java, Swing, Oracle, Java 8** | Designed a database-driven desktop application to manage art store operations, including admin login, art inventory & genre management, employee records, order processing, search functionality, and receipt storage & printing. [\[Github Link\]](#)
- * **Virtual Assistant | Python, Tkinter** | Developed an AI-powered desktop virtual assistant with a user-friendly GUI, supporting voice commands, speech recognition, verbal responses, and weather information retrieval. [\[Github Link\]](#)
- * **Personal Portfolio | HTML, CSS, JavaScript** | Developed a responsive personal portfolio website to showcase skills, education, and projects, with sections for experience, skills, social links, and user-friendly navigation. [\[Github Link\]](#)
- * **Crafty | C#, ASP.NET MVC, Microsoft SQL Server** | Developed an e-commerce website for home-crafted items with database integration, enabling product listings, user interactions, and secure order management. [\[Github Link\]](#)
- * **A Weather Application | Java, Android Studio, OpenWeatherMap API** | Developed a multilingual (English & Bengali) Android weather app with Material Design 2 and dark mode, displaying current location weather, detailed forecasts, and search-based location updates using OpenWeatherMap API and a local database.
- * **Movie Recommender System (What To Watch?) | Python, Pandas, Scikit-learn, Streamlit, Heroku** | Developed an end-to-end content-based movie recommendation system using Movies and Credits datasets. Implemented data preprocessing, feature extraction (genres, keywords, cast, crew), and similarity computation to suggest related movies. Deployed the interactive web app on Heroku with a user interface for viewing movie details and recommendations. [\[Github Link\]](#)
- * **Sensor and Camera-Based Dust Collector Robot | Arduino, Ultrasonic Sensors, Camera Module** | This project involves the development of an autonomous dust collector robot equipped with sensors and a camera module for efficient navigation and obstacle detection. Using ultrasonic sensors for path planning and a camera for visual feedback, the robot detects and collects dust automatically while avoiding obstacles in its environment.

AWARDS & ACHIEVEMENTS

- **Board Scholarship:** Talentpool Scholarship in Primary School Certificate (PSC) and stood 15th in the Sub-district.

- **Programming Contest:** Advanced to the final/qualified round in a team-based competitive programming contest. Intra University Programming Contest, Ahsanullah University of Science and Technology (AUST), 2019
- **Software Competition** Received an award for developing a software project in an intra-university competition. Ahsanullah University of Science and Technology (AUST), 2023

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, R, Assembly.
- **Machine Learning & Deep Learning Frameworks:** TensorFlow, PyTorch, Scikit-learn, OpenCV.
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, Google Colab.
- **Scientific Computing:** MATLAB.
- **XR & Application Development:** Unity, Android (Java, Kotlin, XML, Android Studio), JavaFX, Java Swing.
- **Database Management:** MySQL, PostgreSQL, Oracle.
- **Development & Research Tools:** Git, LaTeX, Linux/Ubuntu, VS Code, JetBrains IDEs.
- **Document Preparation & Visualization:** Overleaf (LaTeX), Microsoft Excel, Adobe Illustrator, Canva.
- **Networking Tools:** Cisco Packet Tracer.

REFERENCES

1. **Mr. Nazmus Sakib**
Assistant Professor, Department of CSE
Ahsanullah University Of Science and Technology
Graduate Researcher
Deakin University
2. **Dr. Mohammad Shafiul Alam**
Professor, Department of CSE
Ahsanullah University Of Science and Technology
3. **Khan Md Hasib**
Assistant Professor, Department of CSE
Ahsanullah University Of Science and Technology
Graduate Researcher
University Of New South Wales, Canberra